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SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 7

Differential of Implicit Functions

Example . $y + xy + y^2 = 2$ differentiate w.r.t x

Solution: $\frac{dy}{dx} + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{-y}{1+x+2y}$ □

Example . $x = ye^x$

Solution: $1 = e^x \frac{dy}{dx} + ye^x, \quad e^x \frac{dy}{dx} = 1 - ye^x, \quad \frac{dy}{dx} = (1 - ye^x) e^{-x} = e^{-x} - y$

$$\frac{dy}{dx} = e^{-x} - \frac{x}{e^x} = \frac{1-x}{e^x} = e^{-x} - xe^{-x}$$
□

Example . If $e^x y = \sin x$, show that $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$

Solution: $e^x y + e^x \frac{dy}{dx} = \cos x$





$$\begin{aligned}\frac{dy}{dx} &= \frac{\cos x - ye^x}{e^x} = \frac{\cos x - (\sin x e^{-x})e^x}{e^x} = \frac{\cos x - \sin x}{e^x} \\ \frac{d^2y}{dx^2} &= \frac{e^x(-\sin x - \cos x) - (\cos x - \sin x)e^x}{e^{2x}} \\ \frac{d^2y}{dx^2} &= \frac{-\sin x - \cos x - \cos x + \sin x}{e^x} = \frac{-2\cos x}{e^x} \\ \frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y &= \frac{-2\cos x}{e^x} + 2\left(\frac{\cos x - \sin x}{e^x}\right) + 2\frac{\sin x}{e^x} = 0\end{aligned}$$

□

7.1. Applications of derivatives

7.1.1. The equation of tangent and normal

Find the equation of tangent and normal to the curve

$$3x^2 - 4y^2 = 9 \text{ at } p(x_1, y_1)$$

Find the slope,

$$6x - 8y\frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{3x}{4y}$$

$$\text{The gradient at } p(x_1, y_1), \quad \text{is} \quad \frac{dy}{dx} = \frac{3x_1}{4y_1}$$



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The equation of the tangent at $p(x_1, y_1)$ is

$$\frac{y-y_1}{x-x_1} = \frac{3x_1}{4y_1} \implies 3xx_1 - 3x_1x_1 = 4yy_1 - 4y_1y_1 \text{ or } 3xx_1 - 3x_1^2 = 4yy_1 - 4y_1^2$$

For normal line $m_1m_2 = -1$, therefore normal gradient is $-\frac{4y_1}{3x_1}$

The equation of the normal at $p(x_1, y_1)$ is

$$\frac{y-y_1}{x-x_1} = -\frac{4y_1}{3x_1} \implies 3yx_1 - 3y_1x_1 = -4xy_1 + 4x_1y_1 \text{ or } 3x_1y - 4y_1x$$

More Revision Questions

(a) Show that the equation of the tangent to $x^2 + xy + y = 0$ at (x_1, y_1) is

$$x(2x_1 - y_1) + y(x_1 + 1) + y_1 = 0$$

(b) Show that the equation of the tangent at (x_1, y_1) to the



curve $ax^2 + by^2 + cxy + dx = 0$ is

$$axx_1 + byy_1 + \frac{1}{2}c(xy_1 + yx_1) + \frac{1}{2}d(x + x_1) = 0$$

(c) Find the slope of the curve $x^2y - xy^2 = 2$ at the point $(-1, -2)$

(d) The parametric equations of a curve are $x = 3(2\theta - \sin 3\theta)$,
 $y = 3(1 - \cos 2\theta)$,

find the equations of the tangent and normal curves at the point p , where $\theta = \frac{\pi}{4}$, the point L and M where they meet the y -axis.

Show that the area of the triangle PLM is $\frac{9}{4}(\pi - 2)$

7.1.2. Hyperbolic functions

The hyperbolic cosine and hyperbolic sine of the real number x are derived by $\cosh x$

and $\sinh x$ and are defined in terms of the exponential func-



tions.

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad \text{and} \quad \sinh x = \frac{e^x - e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad x \neq 0$$

$$\sec hx = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$\csc hx = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \quad x \neq 0$$

$$\cosh^2 x + \sinh^2 x = 1$$

$$1 - \tanh^2 x = \sec^2 hx$$

$$\coth^2 x - 1 = \csc^2 hx$$

$$\sinh(x + y) = \sinh x \cosh y +$$

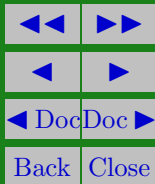
$$\cosh x \sinh y$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sinh 2x =$$

$$2 \sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$





• Derivatives of Hyperbolic functions

i. Let $y = \cosh x = \frac{e^x + e^{-x}}{2}$

then $\frac{dy}{dx} = \frac{e^x - e^{-x}}{2} = \sinh x$

ii. Let $y = \sinh x = \frac{e^x - e^{-x}}{2}$

then $\frac{dy}{dx} = \frac{e^x + e^{-x}}{2} = \cosh x$

iii. Let $y = \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

then $\frac{dy}{dx} = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$ NB $(e^x + e^{-x})(e^x + e^{-x}) =$

$$e^{2x} + 1 + 1 + e^{-2x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{(e^x + e^{-x})^2}$$

$$= \frac{4}{(e^x + e^{-x})^2} = \left(\frac{2}{e^x + e^{-x}}\right)^2 = \sec^2 x$$

iv. Let $y = \coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}} \quad x \neq 0$

then $\frac{dy}{dx} = \frac{(e^x - e^{-x})^2 - (e^x + e^{-x})^2}{(e^x - e^{-x})^2}$

$$= \frac{-4}{(e^x - e^{-x})^2} = -\left(\frac{2}{e^x - e^{-x}}\right)^2 = -\operatorname{csc}^2 x$$



v. Let $y = \sec hx = \frac{2}{e^x + e^{-x}}$

Then $\frac{dy}{dx} = \frac{-2(e^x - e^{-x})}{(e^x + e^{-x})^2} = -\left(\frac{2}{e^x + e^{-x}}\right) \left(\frac{e^x - e^{-x}}{e^x + e^{-x}}\right) = -\sec hx \tanh x$

vi. Let $y = \csc hx = \frac{2}{e^x - e^{-x}}$

Then $\frac{dy}{dx} = \frac{-2(e^x + e^{-x})}{(e^x - e^{-x})^2} = -\left(\frac{2}{e^x - e^{-x}}\right) \left(\frac{e^x + e^{-x}}{e^x - e^{-x}}\right) = -\csc hx \coth x$

Example . $y = \sinh^2 x$

Solution: $\frac{dy}{dx} = 2 \sinh x \cosh x$



Example . $y = \cosh(3x - 1)$

Solution: $\frac{dy}{dx} = 3 \sinh(3x - 1)$



Example . $y = x \sinh x$

Solution: $\frac{dy}{dx} = \sinh x + x \cosh x$





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Example  $y = \ln(\sinh x)$

Solution: $\frac{dy}{dx} = \frac{\cosh x}{\sinh x} = \coth x$

□

EXERCISE 1.  $y = e^{\cosh x}$

EXERCISE 2.  $y = \ln(\sinh x + \cosh x)$

EXERCISE 3.  $y = \tan^{-1}(\tanh x)$

More Revision Questions

Find the derivatives of the following functions

- (1) $f(x) = x^2 \tanh\left(\frac{1}{x}\right)$
- (2) $f(x) = \sec h e^{2x}$
- (3) $f(x) = \sin(\sinh x)$
- (4) $f(x) = 2x \cos^2 x - 2x \sinh^2 x$



Solutions to Exercises

Exercise 1. $\frac{dy}{dx} = \sinh x e^{\cosh x}$

Exercise 1



Exercise 2. $\frac{dy}{dx} = \frac{\sinh x + \cosh x}{\cosh x + \sinh x} = 1$



Exercise 3. then $\tan y = \tanh x$,

$$\frac{dy}{dx} = \frac{\sec h^2 x}{\sec^2 y} = \frac{\sec h^2 x}{\sqrt{1+(\tanh)^2}}$$

$$\sec^2 y \frac{dy}{dx} = \sec h^2 x$$

Exercise 3